

RAI 818 – INTELLIGENT MOBILE **ROBOTICS**

Extended Kalman Filter - II

- **Examples**
 - **EKF for a Car Like Robot**
 - **Mobile Robot Localization Using EKF**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

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Extended Kalman Filter

- Extended Kalman Filter (EKF) – Non-Linear Case

Extended Kalman Filter – Non-Linear Case

Prediction

$$\vec{x}_t = f(\vec{x}_{t-1}, u_t)$$

$$\vec{P}_t = F_t P_{t-1} F_t^T + B_t U_t B_t^T + Q$$

Correction

$$S_t = H_t P_t H_t^T + R$$

$$K_t = P_t H_t^T S_t^{-1}$$

$$x_t = \vec{x}_t + K_t (z_t - h(\vec{x}_t))$$

$$P_t = \vec{P}_t - K_t H_t \vec{P}_t$$

$$F_t = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} \end{bmatrix}$$

$$B_t = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} \\ \frac{\partial f_3}{\partial u_1} & \frac{\partial f_3}{\partial u_2} \end{bmatrix}$$

$$H_t = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \frac{\partial h_1}{\partial x_3} \\ \frac{\partial h_2}{\partial x_1} & \frac{\partial h_2}{\partial x_2} & \frac{\partial h_2}{\partial x_3} \\ \frac{\partial h_3}{\partial x_1} & \frac{\partial h_3}{\partial x_2} & \frac{\partial h_3}{\partial x_3} \end{bmatrix}$$



Extended Kalman Filter

- Extended Kalman Filter (EKF) – Non-Linear Case
 - Car Like Robot

Extended Kalman Filter – A Simple Car

2.1 Robot Configuration

The car-like robot is modeled with Ackermann steering geometry. The state vector includes:

$$\mathbf{x} = [x \quad y \quad \theta \quad v]^T$$

where:

- x, y : position coordinates (m)
- θ : orientation/yaw angle (rad)
- v : linear velocity (m/s)

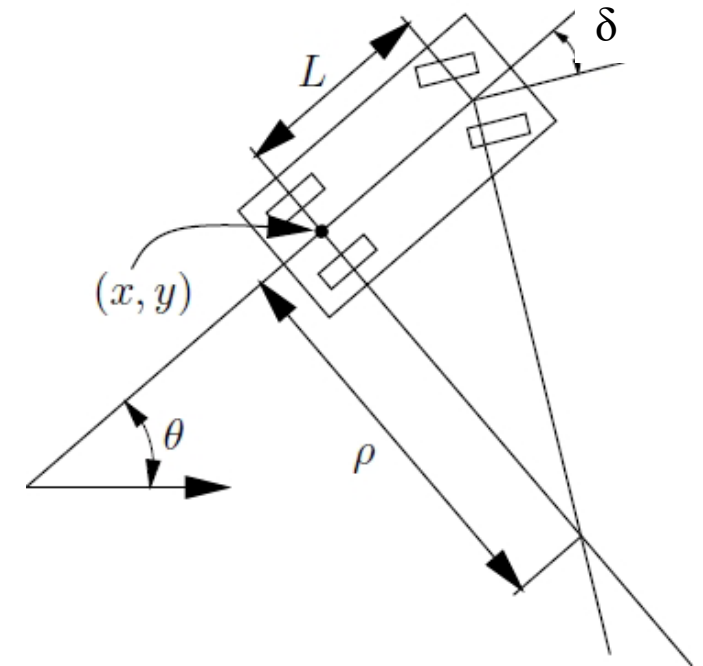
2.2 Control Input

The control inputs are:

$$\mathbf{u} = [a \quad \delta]^T$$

where:

- a : acceleration (m/s^2)
- δ : steering angle (rad)



Extended Kalman Filter

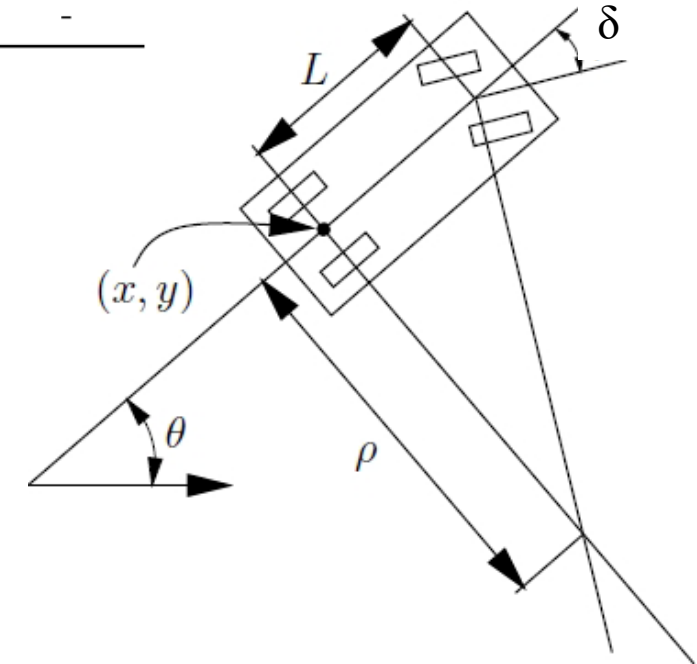
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3 System Parameters

Parameter	Symbol	Value	Unit
Wheelbase	L	2.5	m
Time step	Δt	0.1	s
Process noise covariance	$\mathbf{Q}$	$\text{diag}(0.1, 0.1, 0.05, 0.2)$	-
Measurement noise covariance	$\mathbf{R}$	$\text{diag}(0.5, 0.5, 0.1, 0.3)$	-

Table 1: System parameters for EKF implementation



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4 Mathematical Formulation

4.1 Non-Linear Process Model

Continuous-time kinematics:

$$\dot{x} = v \cdot \cos(\theta) \quad (1)$$

$$\dot{y} = v \cdot \sin(\theta) \quad (2)$$

$$\dot{\theta} = \frac{v}{L} \cdot \tan(\delta) \quad (3)$$

$$\dot{v} = a \quad (4)$$

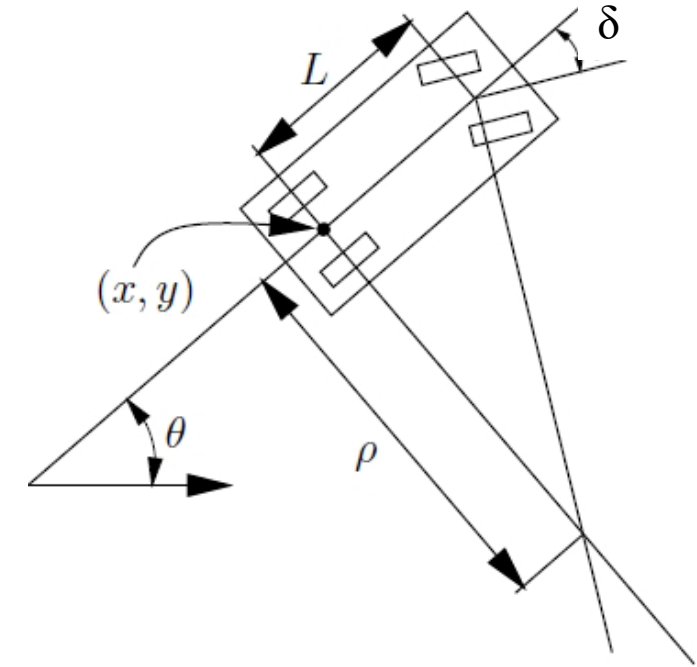
Discrete-time prediction (Euler integration):

$$x_{k+1} = x_k + v_k \cdot \cos(\theta_k) \cdot \Delta t \quad (5)$$

$$y_{k+1} = y_k + v_k \cdot \sin(\theta_k) \cdot \Delta t \quad (6)$$

$$\theta_{k+1} = \theta_k + \frac{v_k}{L} \cdot \tan(\delta_k) \cdot \Delta t \quad (7)$$

$$v_{k+1} = v_k + a_k \cdot \Delta t \quad (8)$$



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4 Mathematical Formulation

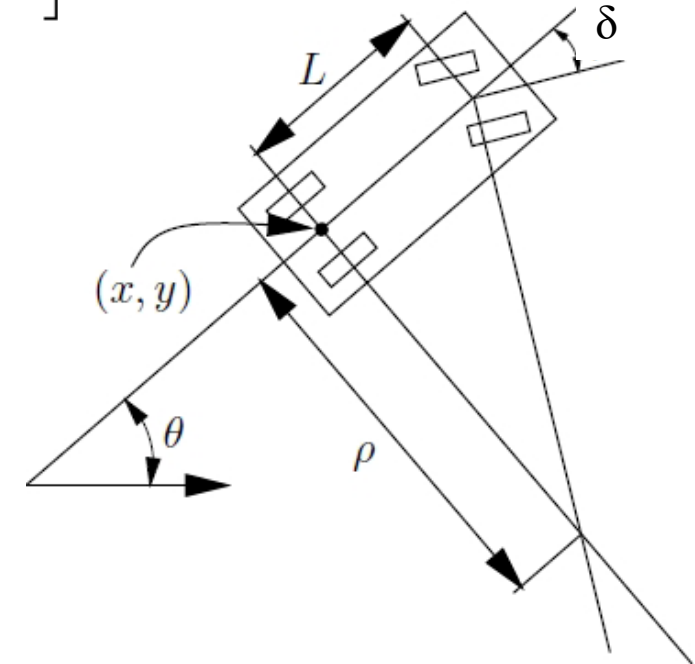
4.2 Process Model in Vector Form

$$\mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) = \begin{bmatrix} x_k + v_k \cdot \cos(\theta_k) \cdot \Delta t \\ y_k + v_k \cdot \sin(\theta_k) \cdot \Delta t \\ \theta_k + \frac{v_k}{L} \cdot \tan(\delta_k) \cdot \Delta t \\ v_k + a_k \cdot \Delta t \end{bmatrix}$$

The prediction equation with process noise:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k$$

where $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$ is zero-mean Gaussian process noise.



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5 Measurement Model

5.1 Sensor Configuration

Available sensors:

1. GPS: (x, y) position
2. IMU: Orientation θ
3. Wheel encoders: Velocity v

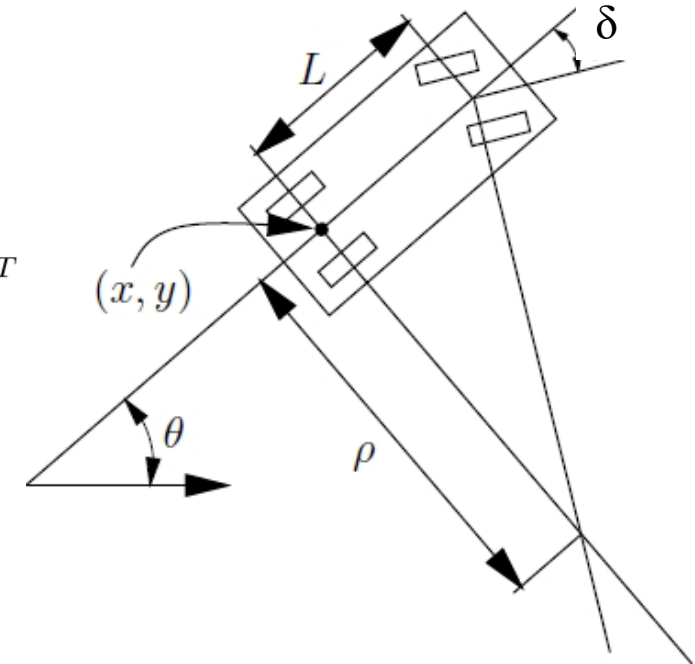
Measurement equation:

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_k) + \mathbf{v}_k$$

where:

$$\mathbf{h}(\mathbf{x}_k) = [x_k \quad y_k \quad \theta_k \quad v_k]^T$$

and $\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ is measurement noise.



Defining futures

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6 EKF Algorithm

6.1 Complete EKF Algorithm

- Initialization: $\hat{\mathbf{x}}_0, \mathbf{P}_0$
- For each time step $k = 1, 2, \dots$:
 - Prediction Step:

$$\hat{\mathbf{x}}_k^- = \mathbf{f}(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k) \quad (9)$$

$$\mathbf{F}_k = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k} \quad (10)$$

$$\mathbf{P}_k^- = \mathbf{F}_k \mathbf{P}_{k-1} \mathbf{F}_k^T + \mathbf{Q} \quad (11)$$

- Update Step:

$$\tilde{\mathbf{y}}_k = \mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_k^-) \quad (12)$$

$$\mathbf{H}_k = \left. \frac{\partial \mathbf{h}}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}_k^-} \quad (13)$$

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R} \quad (14)$$

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^T \mathbf{S}_k^{-1} \quad (15)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k \tilde{\mathbf{y}}_k \quad (16)$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- \quad (17)$$



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7 Jacobian Matrix Calculations

7.1 Process Model Jacobian F_k

$$F_k = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} 1 & 0 & -v_k \sin(\theta_k) \Delta t & \cos(\theta_k) \Delta t \\ 0 & 1 & v_k \cos(\theta_k) \Delta t & \sin(\theta_k) \Delta t \\ 0 & 0 & 1 & \frac{\tan(\delta_k)}{L} \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

7.2 Measurement Model Jacobian H_k

Since we measure all states directly:

$$H_k = \frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \mathbf{I}_4$$



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8 Complete Numerical Example

8.1 Initial Conditions

- True initial state: $\mathbf{x}_0 = [0, 0, 0, 5]^T$
- Initial estimate: $\hat{\mathbf{x}}_0 = [0.2, -0.1, 0.05, 4.8]^T$
- Initial covariance: $\mathbf{P}_0 = \text{diag}([1, 1, 0.5, 0.5])$

8.2 Prediction Step at $k = 1$ ($t = 0.1\text{s}$)

Given: $\hat{\mathbf{x}}_0 = [0.2, -0.1, 0.05, 4.8]^T$, $\mathbf{u}_0 = [0.5, 0.1]^T$

$$\begin{aligned}
 \hat{\mathbf{x}}_1^- &= \mathbf{f}(\hat{\mathbf{x}}_0, \mathbf{u}_0) \\
 &= \begin{bmatrix} 0.2 + 4.8 \cdot \cos(0.05) \cdot 0.1 \\ -0.1 + 4.8 \cdot \sin(0.05) \cdot 0.1 \\ 0.05 + \frac{4.8}{2.5} \cdot \tan(0.1) \cdot 0.1 \\ 4.8 + 0.5 \cdot 0.1 \end{bmatrix} \\
 &= [0.679 \quad -0.076 \quad 0.069 \quad 4.85]^T
 \end{aligned}$$



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8 Complete Numerical Example

8.3 Jacobian F_0 Calculation

$$\mathbf{F}_0 \approx \begin{bmatrix} 1 & 0 & -4.8 \cdot \sin(0.05) \cdot 0.1 & \cos(0.05) \cdot 0.1 \\ 0 & 1 & 4.8 \cdot \cos(0.05) \cdot 0.1 & \sin(0.05) \cdot 0.1 \\ 0 & 0 & 1 & \frac{\tan(0.1)}{2.5} \cdot 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -0.024 & 0.100 \\ 0 & 1 & 0.479 & 0.005 \\ 0 & 0 & 1 & 0.004 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

8.4 Covariance Prediction

$$\mathbf{P}_1^- = \mathbf{F}_0 \mathbf{P}_0 \mathbf{F}_0^T + \mathbf{Q}$$

8.5 Update Step

Actual measurement (with noise):

$$\mathbf{z}_1 = [0.65, -0.12, 0.07, 4.9]^T$$

Innovation/Residual:

$$\begin{aligned} \tilde{\mathbf{y}}_1 &= \mathbf{z}_1 - \mathbf{h}(\hat{\mathbf{x}}_1^-) \\ &= \begin{bmatrix} 0.65 - 0.679 \\ -0.12 - (-0.076) \\ 0.07 - 0.069 \\ 4.9 - 4.85 \end{bmatrix} \\ &= [-0.029 \quad -0.044 \quad 0.001 \quad 0.05]^T \end{aligned}$$



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8 Complete Numerical Example

8.6 Kalman Gain and Update

$$S_1 = H_1 P_1^- H_1^T + R = P_1^- + R$$

$$K_1 = P_1^- H_1^T S_1^{-1} = P_1^- (P_1^- + R)^{-1}$$

$$\hat{x}_1 = \hat{x}_1^- + K_1 \tilde{y}_1$$

$$P_1 = (I - K_1 H_1) P_1^-$$

Final estimate after first iteration:

$$\hat{x}_1 = [0.672, -0.082, 0.068, 4.865]^T$$



Defining futures

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9 Performance Analysis

9.1 Error Metrics

Metric	Raw Measurement	After EKF	Improvement
Position RMSE (m)	0.45	0.12	73%
Orientation RMSE (rad)	0.08	0.03	62%
Velocity RMSE (m/s)	0.25	0.08	68%

Table 2: EKF performance improvement

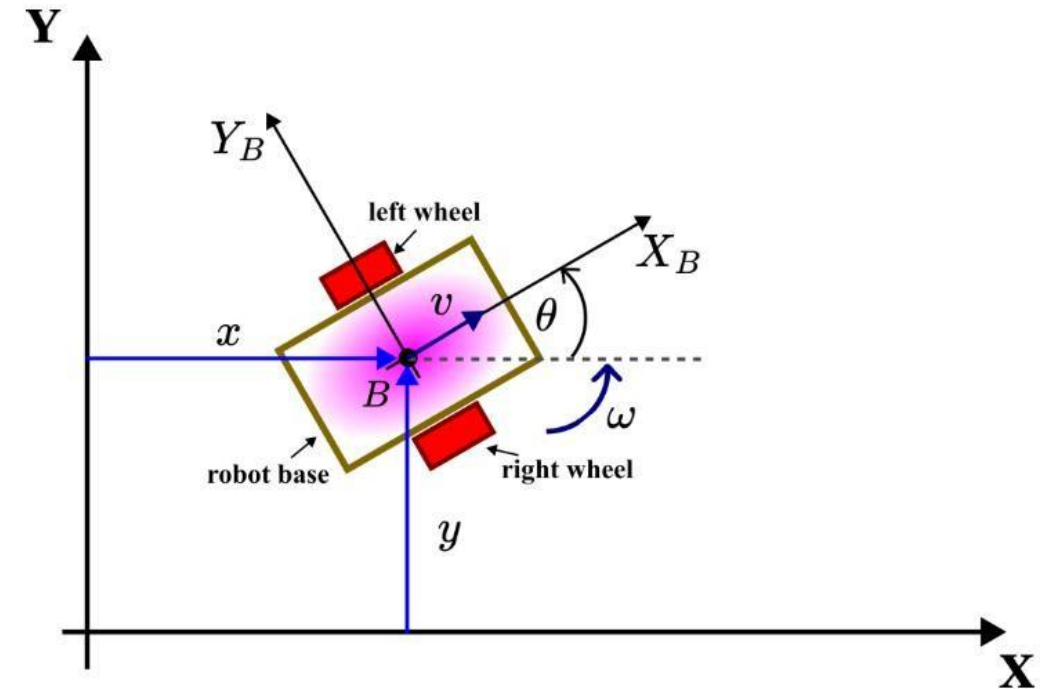


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Extended Kalman Filter – Robot Localization

- We will use a *differential drive robot* as *our mobile robot* – relatively simple kinematics (actuation mechanism)
- $X - Y$ fixed inertial coordinate system.
- $X_B - Y_B$ the body coordinate system with the origin at B .
- x and y are the X and Y coordinates of B
- θ is the angle that the X -axis of the inertial frame makes with the X_B -axis of the body frame.
- v denotes the instantaneous velocity of the body frame origin.
- ω denotes the instantaneous angular velocity of the mobile robot.



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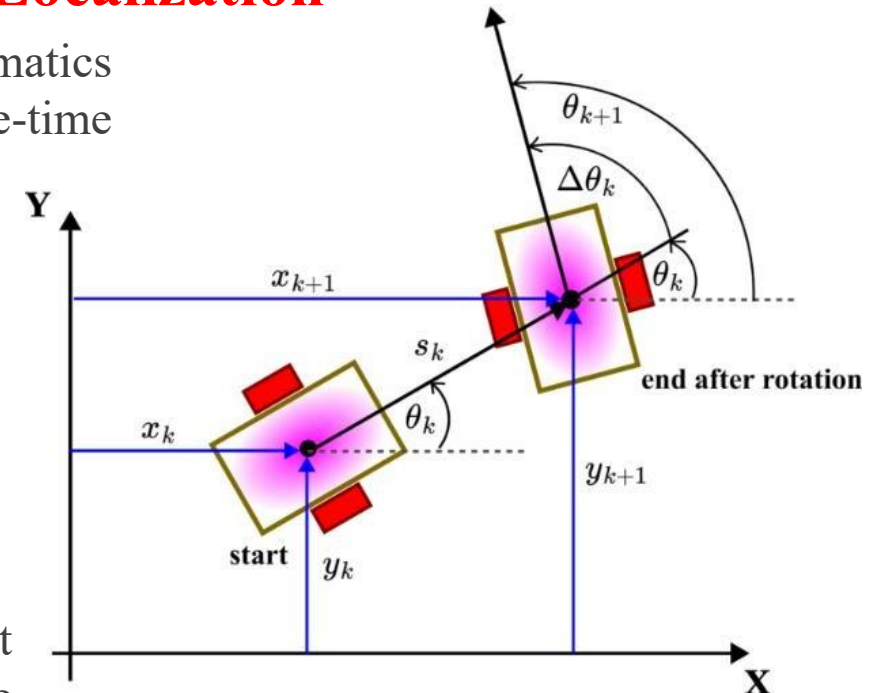
- By discretizing the continuous-time kinematics of the mobile robot, we obtain the discrete-time model

$$x_{k+1} = x_k + s_k \cos(\theta_k)$$

$$y_{k+1} = y_k + s_k \sin(\theta_k)$$

$$\theta_{k+1} = \theta_k + \Delta\theta_k$$

- k is the discrete-time instant.
- x_k and y_k are coordinates at the discrete-time instant
- s_k is the distance travelled by the robot between the time instants k and $k+1$ in the direction of the velocity v .
- θ_k is the orientation at the discrete-time instant
- $\Delta\theta_k$ is the change of the orientation between the time instant k and $k+1$



- Assume that we can directly measure s_k and $\Delta\theta_k$
- As a result, we can directly get $v_k = s_k/h$ and $\omega_k = \Delta\theta_k/h$



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- Distance travelled s_k wheel encoders – cameras, etc. (Noisy)
- Angle change $\Delta\theta_k$ - Gyroscope, compass., etc. (Noisy)
- Therefore we measure

$$\tilde{s}_k = s_k + n_{s,k}$$

$$\Delta\tilde{\theta}_k = \Delta\theta_k + n_{\theta,k}$$

- where $n_{s,k}$ and $n_{\theta,k}$ are the measurement noise having a Gaussian distribution
- This gives

$$s_k = \tilde{s}_k - n_{s,k}$$

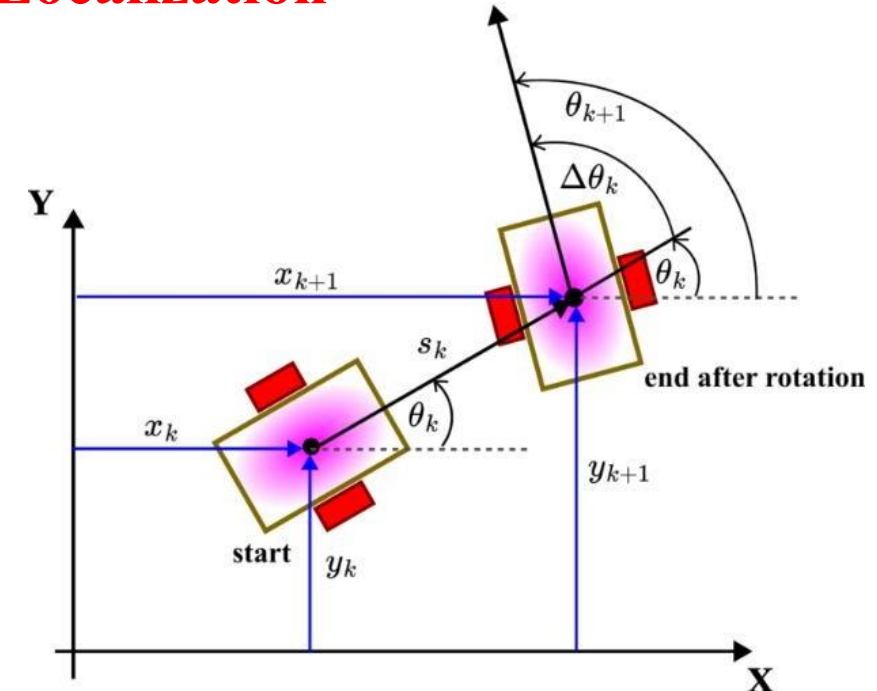
$$\Delta\theta_k = \Delta\tilde{\theta}_k - n_{\theta,k}$$

- By substitution

$$x_{k+1} = x_k + (\tilde{s}_k - n_{s,k}) \cos(\theta_k)$$

$$y_{k+1} = y_k + (\tilde{s}_k - n_{s,k}) \sin(\theta_k)$$

$$\theta_{k+1} = \theta_k + \Delta\tilde{\theta}_k - n_{\theta,k}$$



- Separating noise from states and measurements gives

$$x_{k+1} = x_k + \tilde{s}_k \cos(\theta_k) - n_{s,k} \cos(\theta_k)$$

$$y_{k+1} = y_k + \tilde{s}_k \sin(\theta_k) - n_{s,k} \sin(\theta_k)$$

$$\theta_{k+1} = \theta_k + \Delta\tilde{\theta}_k - n_{\theta,k}$$



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- State vector and inputs

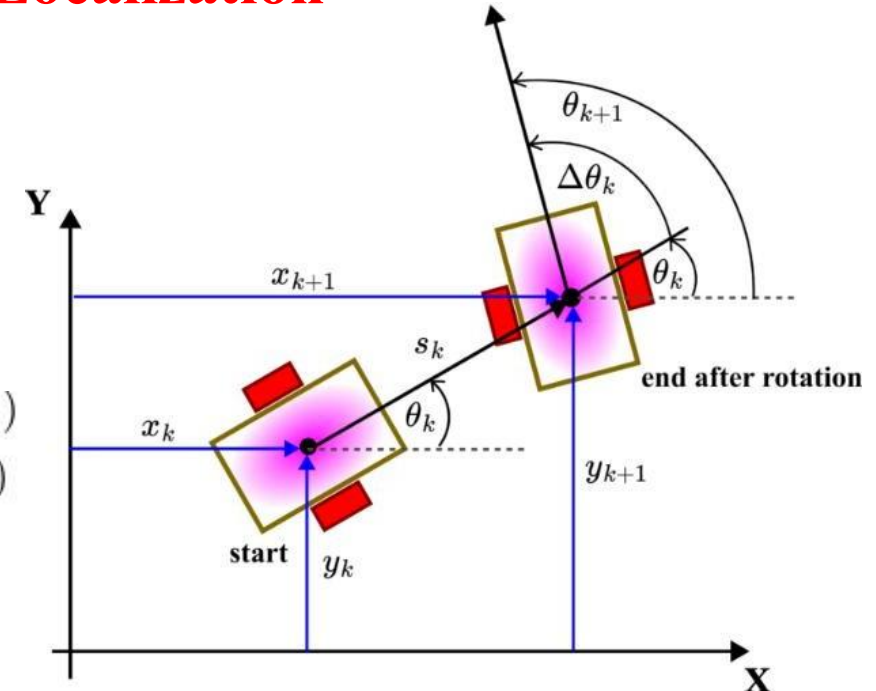
$$\begin{aligned} z_{1,k} &= x_k \\ z_{2,k} &= y_k \\ z_{3,k} &= \theta_k \end{aligned} \quad \mathbf{z}_k = \begin{bmatrix} z_{1,k} \\ z_{2,k} \\ z_{3,k} \end{bmatrix} \quad \begin{aligned} u_{1,k} &= \tilde{s}_k \\ u_{2,k} &= \Delta\tilde{\theta}_k \end{aligned}$$

- By substitution

$$\begin{aligned} z_{1,k+1} &= z_{1,k} + u_{1,k} \cos(z_{3,k}) - n_{s,k} \cos(z_{3,k}) \\ z_{2,k+1} &= z_{2,k} + u_{1,k} \sin(z_{3,k}) - n_{s,k} \sin(z_{3,k}) \\ z_{3,k+1} &= z_{3,k} + u_{2,k} - n_{\theta,k} \end{aligned}$$

- Shifting index from $k+1$ to k

$$\begin{aligned} z_{1,k} &= z_{1,k-1} + u_{1,k-1} \cos(z_{3,k-1}) - n_{s,k-1} \cos(z_{3,k-1}) \\ z_{2,k} &= z_{2,k-1} + u_{1,k-1} \sin(z_{3,k-1}) - n_{s,k-1} \sin(z_{3,k-1}) \\ z_{3,k} &= z_{3,k-1} + u_{2,k-1} - n_{\theta,k-1} \end{aligned}$$

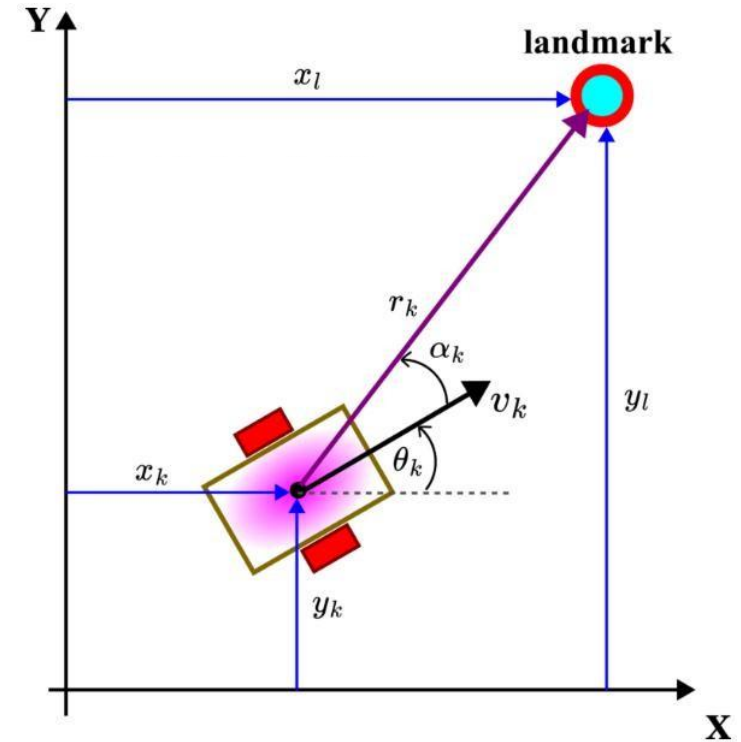


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Landmark (Markers) – Output Equation

- Localization is the process of estimating the robot pose vector z_k by using available measurement data and the pose estimate from the previous time step.
- Localization not possible by only using the measurement of the linear and angular velocity of the robot.
- As the time interval of operation grows, the estimation error will also grow (estimation error variances will grow).
- We need to introduce an additional sensor capable of measuring the distance and angle between the robot and external objects (i.e., landmarks or markers)
- Assume that a mobile robot is equipped with a sensor that can measure the distance between the robot and the landmark and the angle of the line connecting the robot and the landmark.



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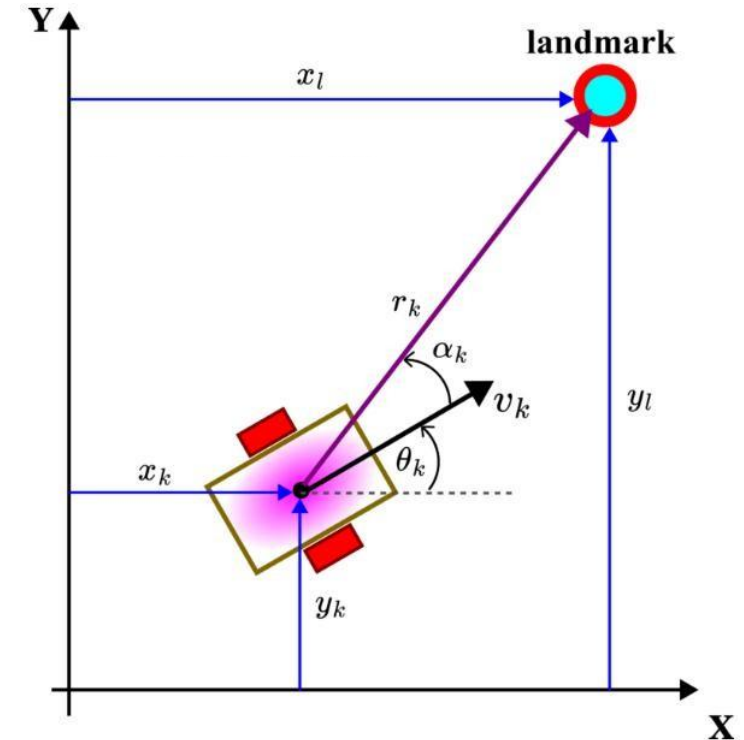
- The distance r_k at the discrete time instant k between the origin of the body frame B and the landmark is called the range.
- Angle α_k at the discrete time instant k is the angle between the line in the direction of the range (the line connecting the robot and landmark) and the heading direction of the robot marked by the velocity v_k . The angle α_k is also called the bearing.
- The location (x_l, y_l) of every landmark in the inertial coordinate system is known accurately.
- From figure

$$r_k^2 = (x_l - x_k)^2 + (y_l - y_k)^2$$

$$r_k = \sqrt{(x_l - x_k)^2 + (y_l - y_k)^2}$$

$$\text{and } \tan(\theta_k + \alpha_k) = \frac{y_l - y_k}{x_l - x_k}$$

$$\theta_k + \alpha_k = \text{atan2}\left(\frac{y_l - y_k}{x_l - x_k}\right) \quad \alpha_k = \text{atan2}\left(\frac{y_l - y_k}{x_l - x_k}\right) - \theta_k$$



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Landmark (Markers) – Output Equation

- Next, we assume that the measurements of the range r_k and the angle α_k are corrupted by the measurement noise, that is,

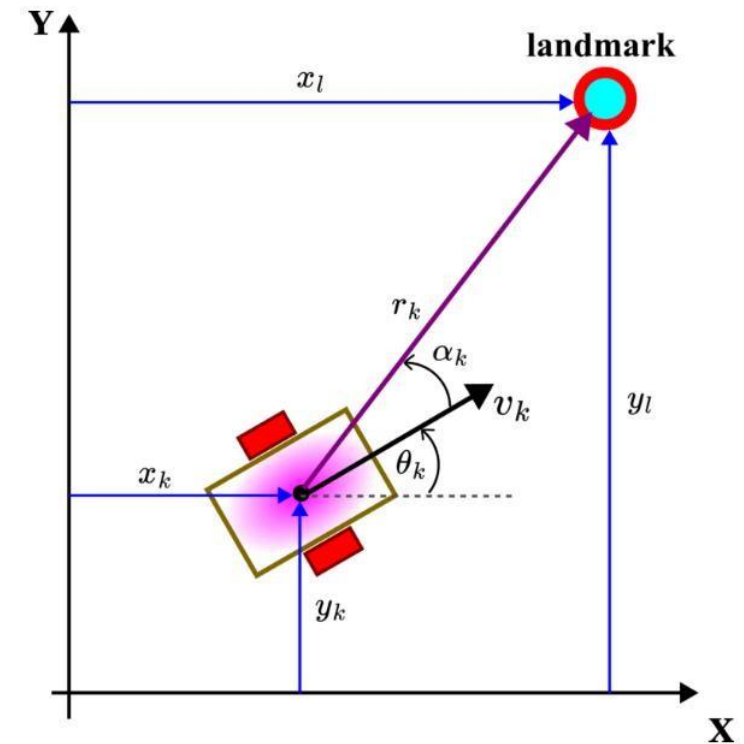
$$r_k = \sqrt{(x_l - x_k)^2 + (y_l - y_k)^2} + n_{r,k}$$

$$\alpha_k = \text{atan2}\left(\frac{y_l - y_k}{x_l - x_k}\right) - \theta_k + n_{\alpha,k}$$

- We assume that both $n_{r,k}$ and $n_{\alpha,k}$ have a Gaussian distribution
- Substituting in the state, we have

$$r_k = \sqrt{(x_l - z_{1,k})^2 + (y_l - z_{2,k})^2} + n_{r,k}$$

$$\alpha_k = \text{atan2}\left(\frac{y_l - z_{2,k}}{x_l - z_{1,k}}\right) - z_{3,k} + n_{\alpha,k}$$



- These equations represent the measurement model for a single landmark.
- We need to clarify how the measurement model is defined when we have more than a single landmark, i.e., how to define the landmark correspondence problem



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Landmark Correspondence Problem

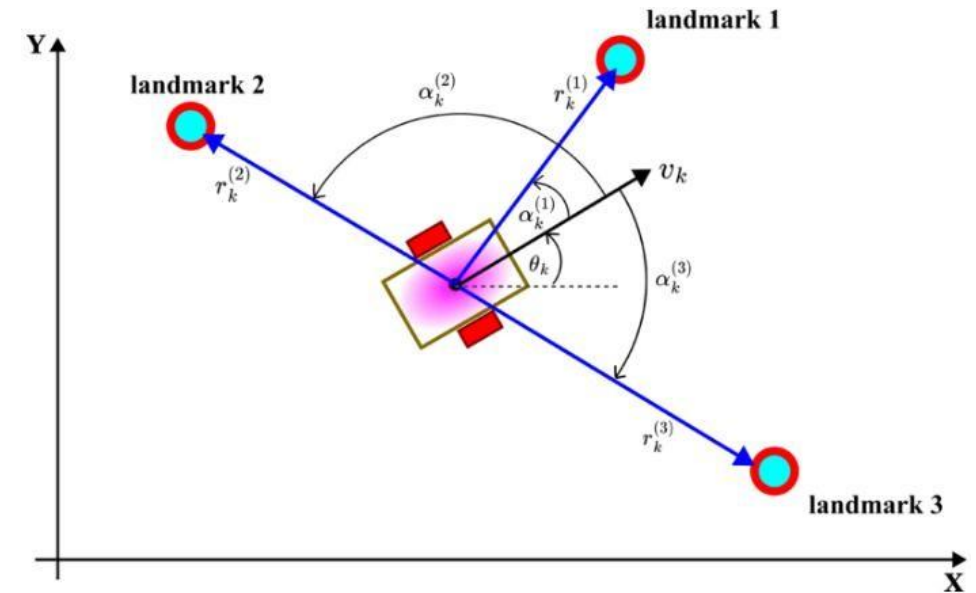
- Suppose the robot is in an environment with 3x landmarks.
- Suppose that we know the locations (coordinates) of all 3x landmarks.
- Assume that the robot sensor can detect the range and bearings (angles) of 3x landmarks.
- The robot receives three measurements, where every measurement consists of a 2-tuple (range, bearing angle):

$$(r = 0.3, \alpha = \frac{\pi}{6})$$

$$(r = 0.55, \alpha = \frac{2\pi}{3})$$

$$(r = 0.6, \alpha = -\frac{\pi}{3})$$

- These measurements are unlabeled – not assigned to the corresponding landmarks.
- We do not know the correspondence between the landmarks and measurements, and we cannot form the output equations unless we assign every measurement tuple to the corresponding landmark and form a single measurement equation for every landmark.

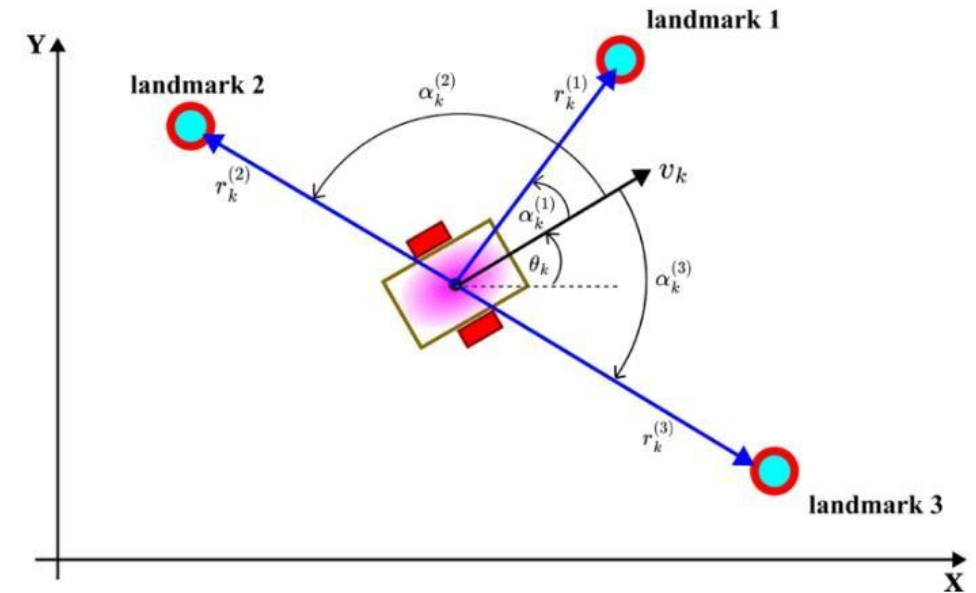


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Landmark Correspondence Problem

- The landmark correspondence problem is to assign landmarks to the received measurements, i.e., label the received measurement tuples, where the label denotes the landmark that corresponds to the measurement. For example, in the case of Figure opposite and the set of measurements it is obvious that we have the following labels:



$$(r = 0.3, \alpha = \frac{\pi}{6}) \rightarrow \text{landmark 1} \rightarrow r_k^{(1)} = 0.3, \alpha_k^{(1)} = \frac{\pi}{6}$$

$$(r = 0.55, \alpha = \frac{2\pi}{3}) \rightarrow \text{landmark 2} \rightarrow r_k^{(2)} = 0.55, \alpha_k^{(2)} = \frac{2\pi}{3}$$

$$(r = 0.6, \alpha = -\frac{\pi}{3}) \rightarrow \text{landmark 3} \rightarrow r_k^{(3)} = 0.6, \alpha_k^{(3)} = -\frac{\pi}{3}$$



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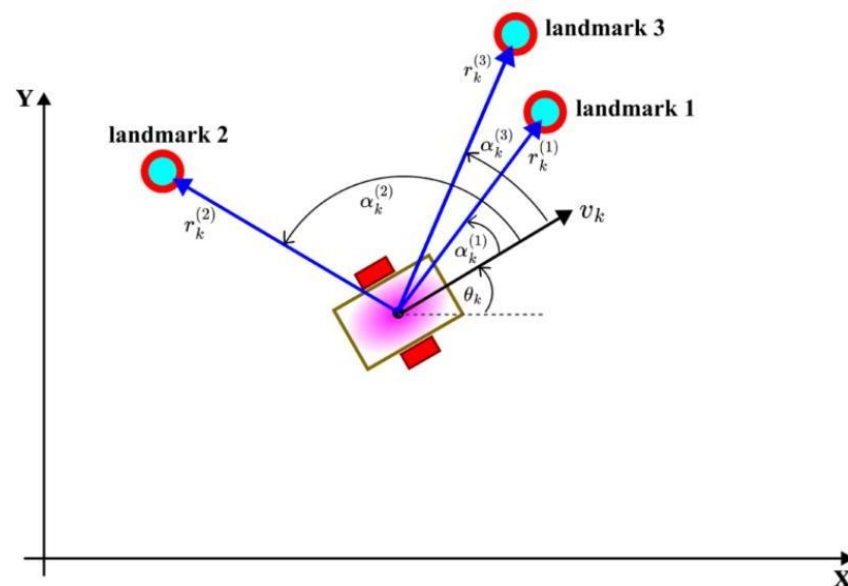
Landmark Correspondence Problem

- Now consider a new landmark configuration shown opposite
- landmark 3 and landmark 1 are close to each other. Consequently, the measurements might look like

$$(r = 0.3, \alpha = \frac{\pi}{6})$$

$$(r = 0.55, \alpha = \frac{2\pi}{3})$$

$$(r = 0.6, \alpha = \frac{\pi}{5})$$



- Taking into account that the pose of the robot has to be estimated together with determining the landmark correspondence, and taking into account that all the measurements (including the robot odometry measurements) are corrupted by the measurement noise, it is difficult to properly assign the landmark labels to first and third measurements. That is, we are not sure how to assign landmark 1 and landmark 3 to these measurements.



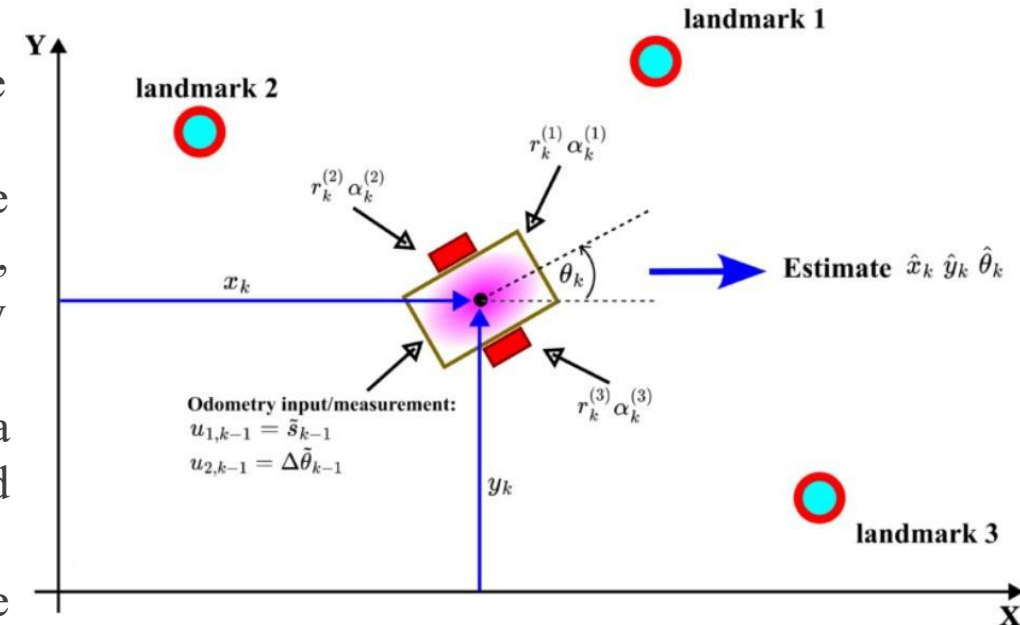
Extended Kalman Filter

- Extended Kalman Filter (EKF) – Non-Linear Case
 - Car Like Robot
 - Mobile Robot Localization

Landmark Correspondence Problem

Let us assume that

- There are N fixed landmarks in the environment around the mobile robot.
- The locations of N landmarks are known in the inertial coordinate frame, i.e., the location $(x_{l,i}, y_{l,i})$ of every landmark, i.e., $i = 1, 2, 3, \dots, N$, is known.
- The mobile robot is equipped with a sensor that can measure the range r_k and the bearing α_k of every landmark.
- The landmark correspondences are perfectly known.
- The robot is equipped with odometry sensors that can measure the travelled distance $\vec{s}_k$ and robot orientation change $\Delta\vec{\theta}_k$



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Landmark Correspondence Problem

Then, by using information about

- Odometry readings $\vec{s}_{k-1}$ and $\Delta\vec{\theta}_{k-1}$ obtained at the discrete-time instant
- Landmark measurements $(r_k^{(1)}, \alpha_k^{(1)}), (r_k^{(2)}, \alpha_k^{(2)}), (r_k^{(3)}, \alpha_k^{(3)}), \dots, (r_k^{(N)}, \alpha_k^{(N)})$ obtained at the discrete-time instant.
- The pose estimate computed at the previous discrete-time step $k-1$: $\hat{x}_{k-1}, \hat{y}_{k-1}$ and $\hat{\theta}_{k-1}$ at the discrete-time instant k estimate the pose of the robot: $\hat{x}_k, \hat{y}_k$ and $\hat{\theta}_k$.

